

Nicolas Von Muhlinen

Fullstack Developer · IoT, Dashboards

Cordoba, Argentina ·

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About Me

Fullstack Developer and Systems Analyst with 4+ years of production experience, specialized in industrial IoT platforms, advanced dashboards, and business rule engines on ThingsBoard. I have delivered mission-critical solutions for airports, data centers, asset management, aquifer monitoring, emissions control, and predictive maintenance, integrating analytics, Machine Learning, and Digital Twins for clients across Latin America and the U.S. I actively incorporate AI tools (Cursor, Claude, Gemini) as a core part of my development workflow. My next professional milestone is specializing in Machine Learning applied to IIoT ecosystems: from raw data to predictive models.

Experience

Fullstack Developer & IoT Specialist | **Senzary**

Aug. 2023 – Present · Miami, FL · Remote

- Design and development of custom dashboards and tailored widgets in ThingsBoard for industrial clients across Argentina, Chile, Ecuador, Colombia, Mexico, and the U.S.
- Delivery of high-impact IoT solutions for airports, data centers, asset management, aquifer monitoring, predictive maintenance, and ESG/climate emissions control.
- Development of advanced analytics, Machine Learning integrations, and Digital Twins for industrial equipment monitoring and fault prediction (Machine Health, IIoT).
- Configuration of ThingsBoard Rule Engines: alarm automation, GPIO control, dust control, PLC integration, and OEM equipment connectivity.
- Integration of SaaS/PaaS platforms, cloud-based IoT data management, and support for Power Management, Electrical Audit, and Facilities Management solutions.
- Frontend development with React, TypeScript, and Angular; state management with Redux Toolkit, REST API and WebSocket consumption for real-time telemetry.
- Technical leadership in dashboard architecture design, cross-functional team coordination, and direct requirements gathering with international clients.
- Unit and integration testing, active participation in code reviews, and Agile/Scrum methodologies in distributed teams.

Featured Projects

Electoral Software Ecosystem

2024– Present · Full-Stack Technical Lead

- Suite of mission-critical web applications for provincial electoral process management.
- Platform for registration, training, and assignment of over 20,000 electoral monitors across Cordoba Capital.
- Vote-counting center system for early election results with real-time data processing and high-concurrency support on election night.
- Institutional websites and public-facing interfaces for the party. Zero downtime on election day.

Stack: React · Next.js · TypeScript · Redux Toolkit · React Hook Form · Zod · Node.js · JWT · Vercel

<https://fiscalizacion2025.vercel.app>

<https://gabrielbornoroni.com.ar/>

IoT B2B Platforms — Chile & Ecuador

2024–2026 · Independent / Senzary

- Advanced analytics dashboards and Rule Engine configuration in ThingsBoard for industrial companies in Chile and Ecuador.
- Direct client requirements gathering, alarm definition, workflow automation, and end-to-end solution delivery.

Stack: ThingsBoard · Rule Engine · MQTT · LoRaWAN · JavaScript · REST APIs

Posada Dormis — Booking Management System

2024 · Thesis Project · Colegio Universitario IES

- Full-stack booking system for a guesthouse: real-time availability, occupancy calendar, and automatic email confirmations.
- Owner administration panel with real-time room status visualization.

Stack: React · Next.js · TypeScript · Node.js · Express · Supabase · PostgreSQL · Nodemailer · Vercel

<https://posada-dormis-frontend-navy.vercel.app/>

Technical Skills

Languages: JavaScript (ES6+), TypeScript

Frontend: React.js, Next.js, Angular, Redux / Redux Toolkit, Zustand, React Hook Form, Zod, TailwindCSS, CSS Modules, HTML5

Backend: Node.js, Express.js, REST APIs, WebSockets

Databases: PostgreSQL, Supabase, Sequelize, MongoDB, Mongoose

Testing: Jest, React Testing Library — unit & integration testing

IoT / Platforms: ThingsBoard (Dashboards + Rule Engine + Custom Widgets), LoRaWAN, MQTT, PLC Integration, GPIO, Digital Twins, IIoT

Industries / Domains: Airports, Data Centers, Aquifer Monitoring, Power Management, Predictive Maintenance, ESG & Climate Control, Asset Management, Facilities Management

Analytics & ML: Machine Learning applied to IoT, predictive analytics, Machine Health, ML model integration in industrial dashboards

DevOps / Tools: Git, Git Flow, GitHub, Vercel, Railway, AWS (intro), SaaS/PaaS cloud

AI-Augmented Development: Cursor, Claude, Gemini — workflow automation, technical prompt engineering

Methodologies: Scrum, Agile, technical leadership, code reviews, pair programming, distributed team collaboration

Education

Systems Analyst & Software Development

Colegio Universitario IES · Feb. 2022 – Mar. 2026 (graduating soon)

Full-Stack Development Bootcamp — 800+ hours

Henry Bootcamp · Nov. 2022 – Apr. 2023

Certifications

- IoT Master Course – Connect to the Future (Udemy, Oct. 2023)
- Introduction to AWS (Udemy, Aug. 2023)
- First Certificate in English – B2 (Dec. 2017)

Languages

Spanish: Native

English: Full professional proficiency (B2)

Portuguese: Basic professional proficiency